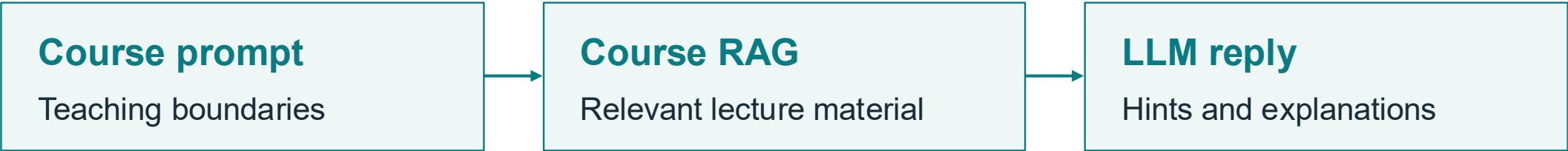


# Background: three ways to organise AI tutoring

Shared problem: giving a correct answer does not show that the student has learned.

## Harvard CS50 Duck

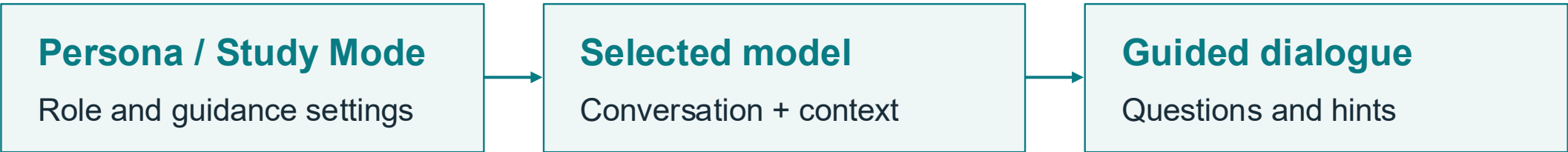
“Why does my loop never stop?”



Published improvement: examples, fine-tuning and human feedback. [1]

## UvA AI Chat

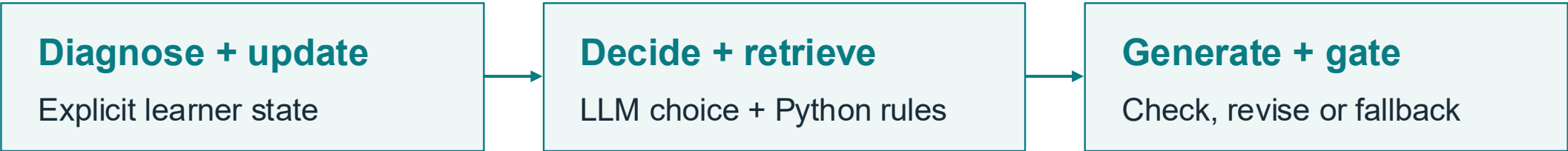
“Quiz me on this topic.”



Configurable teaching interaction within an institutional platform. [2,3]

## Medical AI Tutor

“What does normalization mean?”



Inspected implementation: stateful workflow and external control, using prompts throughout.

# Requirements: MoSCoW priorities

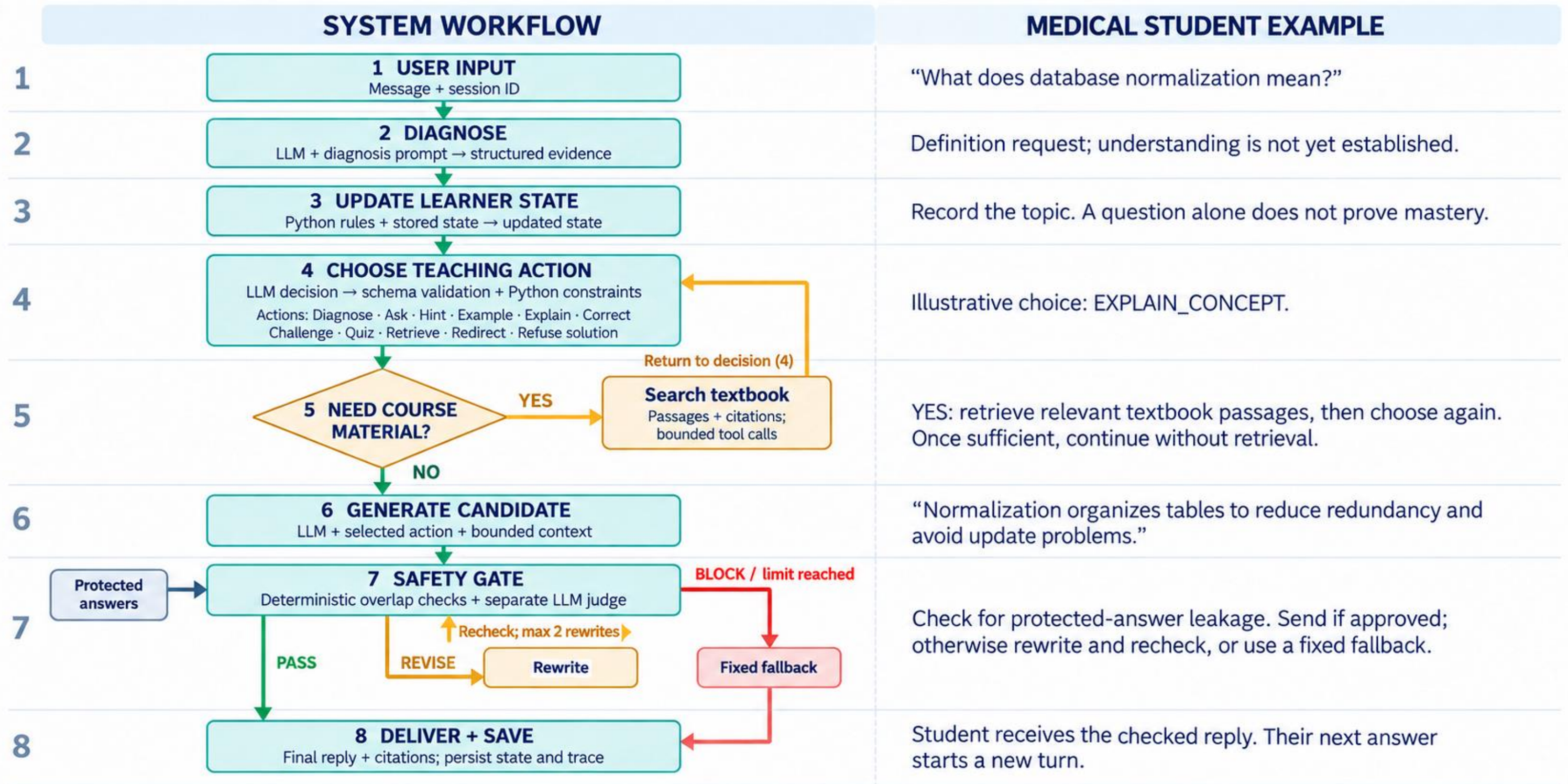
Priority describes importance. “Implemented” describes code, not proven effectiveness.

| Priority             | Requirement and reason                                                                                   | Current position                                                                                        |
|----------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| MUST                 | Guide reasoning; protect official answers.<br>Minimise personal data; use approved processing.           | Teaching workflow and answer gate implemented.<br>Input privacy filter and provider governance needed.  |
| SHOULD               | Keep useful learner state and source citations.<br>Make cost and data handling visible.                  | Local SQLite, bounded context and citations exist.<br>Full call accounting and retention policy needed. |
| COULD                | Support bilingual explanations and a tested local model.<br>Improve accessibility and deployment choice. | Proposed enhancements; not validated features.                                                          |
| WON'T (this version) | Accept real patient records or assign formal grades.<br>Avoid high-stakes use beyond current evidence.   | Proposed operating scope; patient-data blocking and institutional deployment controls still needed.     |

**Local storage is one control. It does not guarantee zero disclosure or establish GDPR compliance.**

# Medical AI Tutor: From Question to Guided Response

Illustrative walkthrough • Each row links a system operation to a student-facing example



Teaching behavior uses prompts; Python controls state, execution limits and validation. Example decisions are illustrative, not a recorded run.

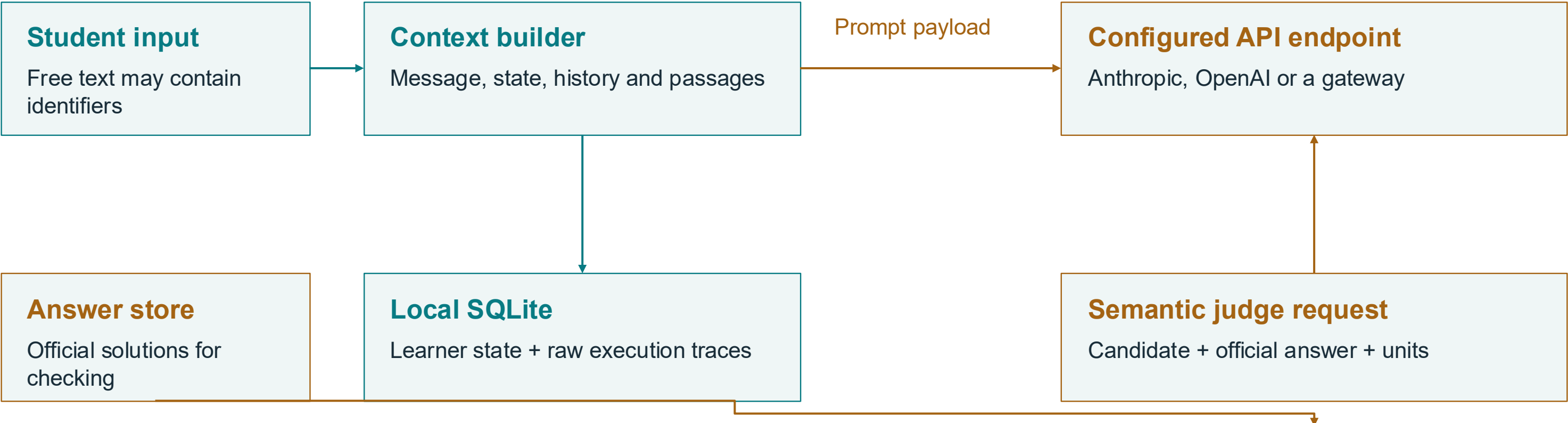
Code clarification: protected-answer checks run only when a protected exercise is active. This is an illustrative walkthrough.

# User input: where the data actually goes

A local Python application can still send personal data to an external model service.

## APPLICATION HOST

## EXTERNAL PROCESSING



**No verified general personal-data scrubber before these calls.**

Review all recipients. GDPR Arts. 13, 28, 44–46. [6–8]

# Input risks and the controls they require

The answer-leakage gate protects the exercise solution; personal-data protection needs separate controls.

| Possible input / exposure                                         | Required response                                                                                      | Legal support                                            |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| “Patient A, born ..., has ...”<br>Identifiable health information | Use synthetic cases. Block or remove identifiers before processing; require a valid basis if retained. | Arts. 6 and 9<br>Lawfulness + special-category condition |
| “I failed this topic again ...”<br>Identifiable learner profile   | Explain what is inferred and why.<br>Keep only task-relevant state.                                    | Arts. 5(1)(c) and 13<br>Minimisation + transparency      |
| Repeated history in API calls<br>Unnecessary external disclosure  | Limit fields per role, inspect payloads, record recipients and processor terms.                        | Arts. 5 and 28<br>Minimisation + processing agreement    |
| Shared endpoint or database<br>Unauthorised session access        | Add authentication, session authorisation, secure hosting and risk-based encryption.                   | Art. 32<br>Appropriate security measures                 |

**Current gap: API routes have no authentication layer in the supplied app. A session ID is not access control.**

# Cloud processing: model name is not the data policy

Compare the actual service and contract used by the application, rather than the consumer chatbot brand.

| Service                    | Training and retention                                                                                                                        | Implication for this project                                                                                   |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| OpenAI API                 | No training by default unless opted in.<br>Default abuse logs can be retained up to 30 days, with exceptions and eligible retention controls. | Verify organisation settings, endpoint and DPA.<br>Do not apply ChatGPT consumer settings to API use.          |
| Anthropic API              | No training of retained data without permission.<br>Retention / ZDR eligibility varies by model, feature and agreement.                       | Confirm terms for the exact model and account.<br>Do not assume a universal zero-retention policy.             |
| Azure-hosted OpenAI models | Microsoft hosts inference; prompts are not sent to OpenAI services. Abuse monitoring and processing geography depend on deployment.           | UvA describes approved Azure processing.<br>Its arrangements do not automatically cover this personal project. |

**Before a pilot: identify controller and processors, sign appropriate terms, and assess any non-EEA transfer.**

# Local records, user rights and textbook ownership

Local persistence reduces one dependency, but the operator still needs a complete data lifecycle.

## USER RECORDS

### Implemented

SQLite stores state and traces. Reset deletes session records and recreates an empty session.

### Still needed

Define retention and access controls. Provide access, correction and erasure procedures.

### Deletion boundary

Local reset does not delete provider logs or prove removal from backups and SQLite journals.

## COURSE MATERIAL

### Licensed textbook

Winter et al., Health Information Systems (2023). The project identifies the text as CC BY 4.0.

### Reuse obligations

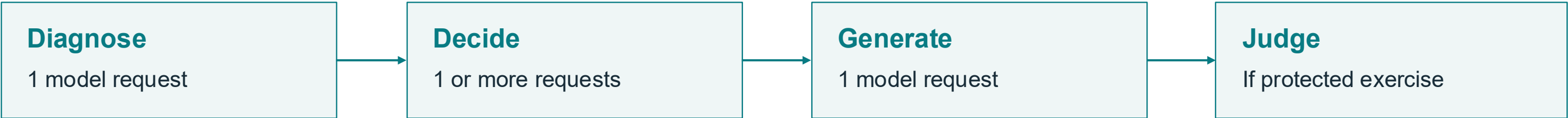
Retain attribution and licence, indicate changes, and check separately credited third-party material.

### Two different protections

Withholding official answers is a teaching rule. Privacy concerns identifiable people. Copyright concerns permission to reuse protected content.

# One student turn can trigger several model calls

A separate generator and judge are only part of the full workflow.



| Path                    | Typical successful requests | Why it changes                                                               |
|-------------------------|-----------------------------|------------------------------------------------------------------------------|
| General question        | 3                           | Diagnosis + decision + generation                                            |
| Protected exercise      | 4                           | Add one semantic judge                                                       |
| Protected + one rewrite | 6                           | Add rewrite and a new judge check                                            |
| Retrieval / retries     | Additional requests         | Each re-decision adds a call; schema repair and fallback checks can add more |

**Accounting gap: the current per-turn counter can omit re-checks and provider-level repairs.**

# Model choice: capacity, task fit and what is unknown

GPT-4o is the code’s OpenAI fallback, not a verified live configuration for this installation.

| Model            | Published capacity                                                              | Role to evaluate                                                               |
|------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| GPT-4o           | 128K context; text/image input<br>Parameter count not disclosed in cited docs   | Baseline for generation and judgment                                           |
| GPT-4o mini      | 128K context; structured outputs<br>Parameter count not disclosed in cited docs | Lower-cost diagnosis / decision candidate                                      |
| Gemini 2.5 Flash | 1,048,576 input-token limit<br>Parameter count not disclosed in cited docs      | Alternative provider comparison; integration needed                            |
| Llama 3.1 8B     | 8 billion parameters; 128K context<br>Deployable model weights                  | Controlled local-hosting experiment;<br>hardware and quality validation needed |

**Performance here means correct diagnosis, appropriate hints, low leakage and useful feedback at acceptable latency.**

# Token cost: an explicit scenario

Assume 1,000 input + 250 output tokens per request; standard uncached text pricing in USD.

## GPT-4o

\$2.50 input / \$10.00 output per million tokens

## GPT-4o mini

\$0.15 input / \$0.60 output per million tokens

| Requests per turn    | GPT-4o / turn | GPT-4o / 1,000 turns | Mini / 1,000 turns |
|----------------------|---------------|----------------------|--------------------|
| 2: simple comparison | \$0.010       | \$10.00              | \$0.60             |
| 3: general path      | \$0.015       | \$15.00              | \$0.90             |
| 4: protected path    | \$0.020       | \$20.00              | \$1.20             |
| 6: one rewrite path  | \$0.030       | \$30.00              | \$1.80             |

**Cost =  $\Sigma$  (input tokens × input rate + output tokens × output rate) / 1,000,000**

# Environmental impact: training and repeated inference

Reduce resource use only after checking that learning support and safety remain adequate.

## INITIAL IMPACT

**No new model training in this project.**

Pretrained models still have upstream training and hardware impacts. The API bill does not provide a reliable allocation of that footprint.

## OPERATIONAL IMPACT

**Every repeated prompt and reply requires inference work.**

Existing controls: bounded dialogue, limited rewrites, conditional retrieval and lexical search. Extra diagnosis and judgment also add work.

---

## COMPARE

**All GPT-4o / mini for controller + GPT-4o for reply and judge / Gemini or local Llama pilot**

Hold cases constant. Measure tokens, latency, safety and learning quality. Measure energy for local runs. API carbon ranking remains unknown without comparable energy and deployment data.

# Bias: the model may misread the learner

Potential risks inferred from the design; these have not been measured in this project.

| Potential source         | Effect on the student                                                                | Mitigation / test to add                                                                     |
|--------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Language and expression  | Broken English or a short answer may be misread as weak conceptual understanding.    | Pair equivalent answers in different language styles; compare diagnosis and teaching action. |
| Persistent learner state | An early wrong inference may keep a student on overly simple or repetitive material. | Let students contest estimates; test recovery after new contradictory evidence.              |
| Textbook and context     | A course-specific example may be treated as universally applicable.                  | Ask for context, show citations and include alternative valid explanations in evaluation.    |
| Model acting as judge    | A separate call may reproduce the same blind spots as the generating model.          | Use human-labelled cases and deterministic checks; measure false refusals as well as leaks.  |

**Human understanding is not directly observable from a fluent answer or a single model score.**

# Ethics and personal reflection

Draft position for the presenter to review; no personal experiences or learning gains are invented.

## What I value

I see value in timely hints, source-linked explanations and a visible teaching process.

## What concerns me

Plausible responses can hide misunderstanding. Helpful scaffolding can also become cognitive dependence.

## My proposed boundary

I would use it for practice, with human oversight and no real patient records or automatic formal grading.

---

## Evidence changes the position

The mock-mode screenshot showed irrelevant repeated hints. A tutoring tone alone did not establish understanding.

A real learning evaluation should include an unaided follow-up task, not just successful assisted answers.

# References: systems and data protection

Official documentation and research, accessed 11 September 2026. Click a source title to open it.

|                                                         |                                                                                                    |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <a href="#"><u>[1] CS50 architecture</u></a>            | Liu et al. (2025). Improving AI in CS50: Leveraging Human Feedback for Better Learning. SIGCSE TS. |
| <a href="#"><u>[2] UvA teaching configuration</u></a>   | UvA. AI Chat user manual, v1.1 (2025), Study Mode and Personas.                                    |
| <a href="#"><u>[3] UvA data processing</u></a>          | UvA. AI Chat en verantwoord gebruik van AI. Institutional account of data flow and system prompts. |
| <a href="#"><u>[4] GDPR principles</u></a>              | EDPB. Data protection basics. GDPR Arts. 4, 5, 6, 9 and 13.                                        |
| <a href="#"><u>[5] GDPR security</u></a>                | EDPB. Secure personal data. Risk-appropriate technical and organisational measures, Art. 32.       |
| <a href="#"><u>[6] GDPR individual rights</u></a>       | EDPB. Respect individuals' rights. Transparency, access, rectification and erasure.                |
| <a href="#"><u>[7] GDPR international transfers</u></a> | EDPB. International data transfers. Adequacy and appropriate safeguards under Chapter V.           |
| <a href="#"><u>[8] GDPR processing partners</u></a>     | EDPB. FAQ. Controller–processor agreements and lawful processing.                                  |
| <a href="#"><u>[9] OpenAI API privacy</u></a>           | OpenAI. Data controls in the OpenAI platform. API training and retention policies.                 |
| <a href="#"><u>[10] Anthropic API privacy</u></a>       | Anthropic. API and data retention. Model-, feature- and agreement-specific conditions.             |
| <a href="#"><u>[11] Azure model hosting</u></a>         | Microsoft. Data, privacy and security for models sold by Azure.                                    |

# References: models, environment and learning

Official documentation and research, accessed 11 September 2026. Click a source title to open it.

[\[12\] Textbook reuse](#)

Creative Commons. Attribution 4.0 International legal code, Section 3.

[\[13\] GPT-4o specification](#)

OpenAI. GPT-4o model documentation and standard text-token pricing.

[\[14\] GPT-4o mini specification](#)

OpenAI. GPT-4o mini model documentation and standard text-token pricing.

[\[15\] Gemini comparison](#)

Google. Gemini 2.5 Flash model documentation.

[\[16\] Llama comparison](#)

Meta. Llama 3.1 model card. Parameter sizes and deployment context.

[\[17\] Inference energy](#)

Luccioni, Jernite & Strubell. Power Hungry Processing: Watts Driving the Cost of AI Deployment?

[\[18\] Efficiency trade-offs](#)

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[\[19\] Language bias](#)

Fleisig et al. (2024). Linguistic Bias in ChatGPT: Language Models Reinforce Dialect Discrimination.

[\[20\] Learning and dependence](#)

Bastani et al. Study discussed by Knowledge at Wharton: Without Guardrails, Generative AI Can Harm Education.

[\[21\] Course knowledge base](#)

Winter et al. (2023). Health Information Systems, third edition. Springer.